Practical no 03 . Basic Layout Management

A) Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

Project Name: LinearLayout
Program Code:
Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">
  <!-- Title -->
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Student Registration" android:textSize="26sp"
     android:textStyle="bold"
     android:layout_gravity="center" android:layout_marginBottom="20dp"/>
  <!--Vertical Layout Components -->
  <EditText
     android:id="@+id/etName"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:hint="Enter Name"
     android:layout_marginBottom="10dp"/>
  <EditText
     android:id="@+id/etEmail"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:hint="Enter Email"
     android:layout_marginBottom="20dp"/>

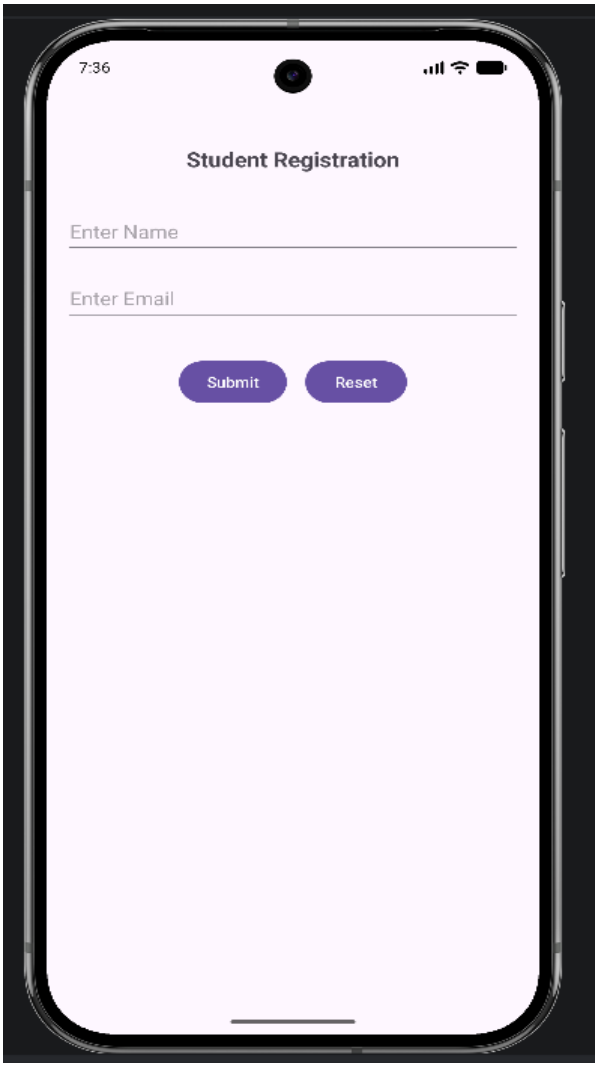
  <!-- Horizontal LinearLayout -->
  <LinearLayout
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:orientation="horizontal"
```

```
    android:gravity="center">

    <Button
      android:id="@+id/btnSubmit"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Submit"/>
    <Button
      android:id="@+id/btnReset"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Reset"
      android:layout_marginStart="20dp"/>
  </LinearLayout>
</LinearLayout>
```

Output:

LinearLayout



B) Develop an Android application using RelativeLayout to align UI components dynamically.

Project Name: RelativeLayout
Program Code:
Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/titleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:text="RelativeLayout Demo"
        android:textAlignment="center"
        android:textSize="20sp"
        android:textStyle="bold" />

    <ImageView
        android:id="@+id/imageView6"
        android:layout_width="268dp"
        android:layout_height="126dp"
        android:layout_below="@id/titleText"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        app:srcCompat="@drawable/zoro" />

    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="match_parent"
        android:layout_height="64dp"
        android:layout_below="@id/imageView6"
        android:layout_marginTop="20dp"
        android:ems="10"
        android:hint="Project Name"
        android:inputType="text" />
```

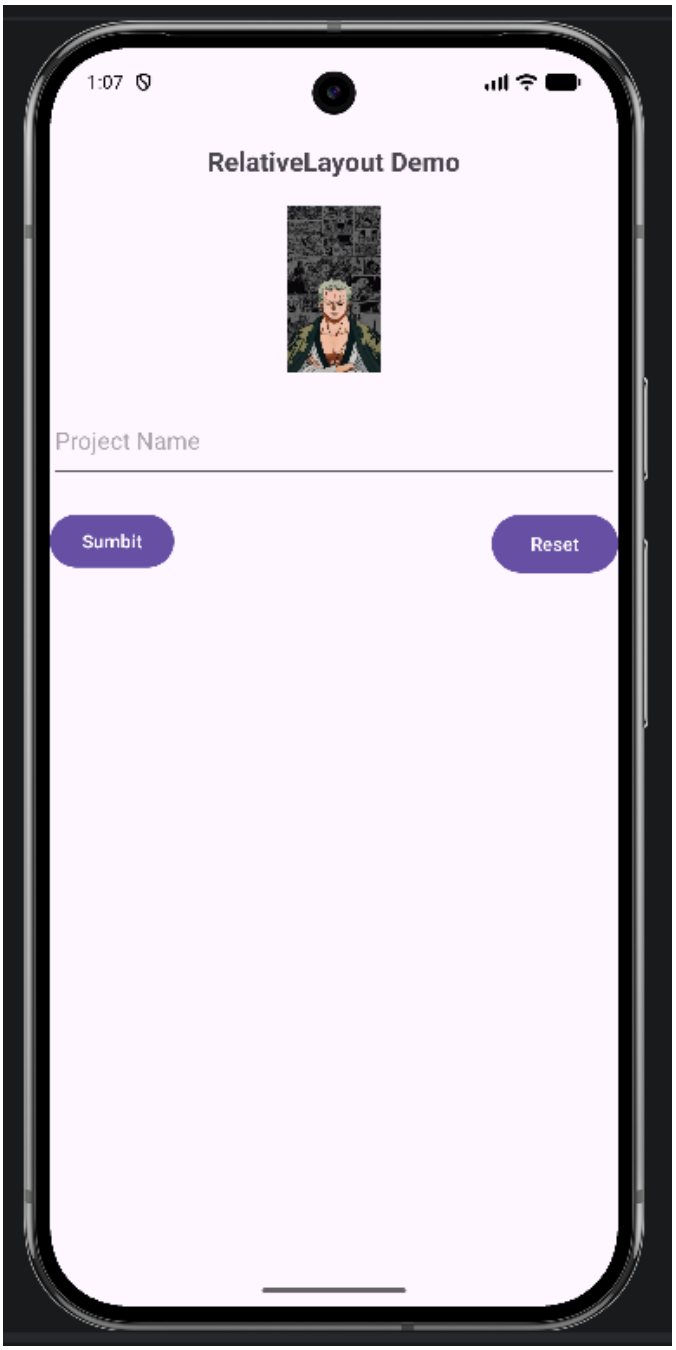
```
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/editTextText2"
        android:layout_alignParentStart="true"
        android:layout_marginTop="20dp"
        android:text="Sumbit" />

    <Button
        android:id="@+id/button2"
        android:layout_width="95dp"
        android:layout_height="52dp"
        android:layout_below="@id/editTextText2"
        android:layout_alignParentEnd="true"
        android:layout_marginTop="20dp"
        android:text="Reset" />

</RelativeLayout>
```

Output:

Relative Layout



C) Develop an Android application using TableLayout to design a structured Login/Registration Form.

Project Name: TableLayout
Program Code:
Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:fillViewport="true"
  tools:context=".MainActivity">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
      android:id="@+id/tvHeader"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginBottom="32dp"
      android:text="Registration Form"
      android:textColor="@android:color/black"
      android:textSize="28sp"
      android:textStyle="bold" />

    <TableLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:stretchColumns="1">

      <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <TextView
          android:layout_width="wrap_content"
          android:layout_height="wrap_content"
          android:layout_gravity="center_vertical"
```

```
            android:paddingEnd="12dp"
            android:text="Name:"
            android:textColor="@android:color/black"
            android:textSize="16sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/etName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Name"
            android:inputType="textPersonName" />
    </TableRow>

    <!-- Email Row -->
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center_vertical"
            android:paddingEnd="12dp"
            android:text="Email:"
            android:textColor="@android:color/black"
            android:textSize="16sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/etEmail"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Email"
            android:inputType="textEmailAddress" />
    </TableRow>

    <!-- Password Row -->
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```
            android:layout_gravity="center_vertical"
            android:paddingEnd="12dp"
            android:text="Password:"
            android:textColor="@android:color/black"
            android:textSize="16sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Password"
            android:inputType="textPassword" />
    </TableRow>

    <!-- Buttons Row -->
    <TableRow
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_span="2"
            android:orientation="horizontal">

            <Button
                android:id="@+id/btnRegister"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_marginEnd="8dp"
                android:layout_weight="1"
                android:backgroundTint="#3F51B5"
                android:text="Register"
                android:textAllCaps="false"
                android:textColor="@android:color/white"
                app:cornerRadius="20dp" />

            <Button
                android:id="@+id/btnReset"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_marginStart="8dp"
                android:layout_weight="1"
                android:backgroundTint="#3F51B5"
                android:text="Reset"
```

```
            android:textAllCaps="false"
            android:textColor="@android:color/white"
            app:cornerRadius="20dp" />
        </LinearLayout>
      </TableRow>
    </TableLayout>
  </LinearLayout>
</ScrollView>
```

Output:

Table Layout

